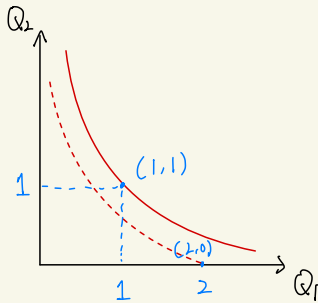


Problem 2

$$U(Q_1, Q_2) = 12(Q_1 + Q_2) - (Q_1^2 + Q_2^2) - Q_1 Q_2$$

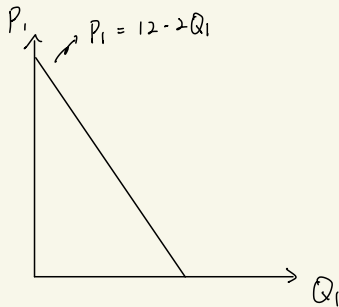


$$U(1,1) > U(2,0) \rightarrow \text{love of variety}$$

1. Monopoly : $Q_2 = 0$

Derive demand for Q_1 at $Q_2 = 0$: $\left. \frac{\partial U}{\partial Q_1} \right|_{Q_2=0} = 12 - 2Q_1 = P_1$

Marginal utility
= Willingness to pay

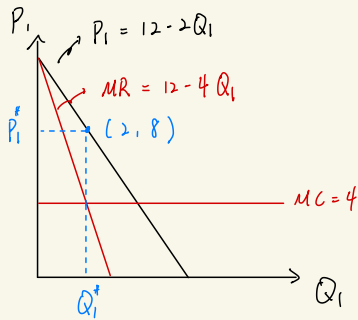


Monopolistic firm problem :

$$\begin{aligned} \pi_1 &= \max_{Q_1} \{ P_1 Q_1 - MC \times Q_1 \} \\ &= \max_{Q_1} \{ (12 - 2Q_1) Q_1 - MC \times Q_1 \} \end{aligned}$$

$$\frac{d\pi_1}{dQ_1} = 12 - 4Q_1 - \underbrace{MC_1}_{=4} \stackrel{\text{first order condition}}{=} 0$$

$$\Rightarrow \underbrace{12 - 4Q_1}_{MR} = \underbrace{4}_{MC} \rightarrow \text{profit maximization condition}$$



$$\Rightarrow Q_1^* = 2 \rightarrow \text{plug in the demand function}$$

$$P_1^* = 8$$

$$\begin{aligned} \pi_1 &= P_1^* Q_1^* - MC_1 \times Q_1^* \\ &= 16 - 8 \\ &= 8 \end{aligned}$$

$$2. \text{ Trade : } Q_2 \neq 0, \quad MC_1 = 4, \quad MC_2 = 1$$

$$\text{Demand for } Q_1 : \frac{\partial U}{\partial Q_1} = 12 - 2Q_1 - Q_2 = P_1$$

$$\text{Demand for } Q_2 : \frac{\partial U}{\partial Q_2} = 12 - 2Q_2 - Q_1 = P_2$$

$$\text{Firm 1's profit : } \pi_1 = \max_{Q_1} \left\{ (12 - 2Q_1 - Q_2) Q_1 - 4Q_1 \right\}$$

$$\frac{\partial \pi_1}{\partial Q_1} = \underbrace{12 - 4Q_1 - Q_2}_{MR_1} - \underbrace{4}_{MC_1} = 0$$

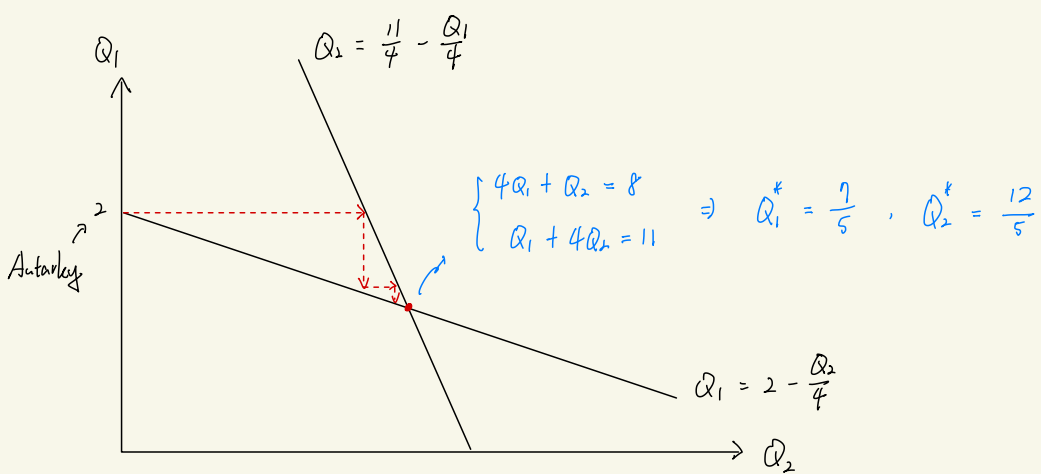
$$\text{Firm 2's profit : } \pi_2 = \max_{Q_2} \left\{ (12 - 2Q_2 - Q_1) Q_2 - Q_2 \right\}$$

$$\frac{\partial \pi_2}{\partial Q_2} = \underbrace{12 - 4Q_2 - Q_1}_{MR_2} - \underbrace{1}_{MC_2} = 0$$

Reaction function :

$$Q_1 = 2 - \frac{Q_2}{4}$$

$$Q_2 = \frac{11}{4} - \frac{Q_1}{4}$$



Plug Q_1^*, Q_2^* into the demand function :

$$P_1^* = 12 - 2 \times \frac{7}{5} - \frac{12}{5} = \frac{34}{5}$$

$$P_2^* = 12 - 2 \times \frac{12}{5} - \frac{7}{5} = \frac{29}{5}$$

$$\pi_1 = P_1^* Q_1^* - 4 Q_1^* = \frac{34}{5} \times \frac{7}{5} - 4 \times \frac{7}{5} = \frac{98}{25}$$

$$\pi_2 = P_2^* Q_2^* - Q_2^* = \frac{29}{5} \times \frac{12}{5} - \frac{12}{5} = \frac{288}{25}$$

3. $CS = U(Q_1, Q_2) - P_1 Q_1 - P_2 Q_2$, $PS = \pi_1$

Before trade : $Q_1^A = 2$, $P_1^A = 8$, $Q_2^A = 0$

$$CS = U(2, 0) - 8 \times 2 = 20 - 16 = \underline{4}$$

$$PS = \pi_1 = \underline{8}$$

$$\text{Welfare} = CS + PS = 4 + 8 = \underline{12}$$

$$\text{After trade : } Q_1^T = \frac{7}{5}, \quad Q_2^T = \frac{12}{5}, \quad P_1^T = \frac{34}{5}, \quad P_2^T = \frac{29}{5}$$

$$CS = U\left(\frac{7}{5}, \frac{12}{5}\right) - \frac{34}{5} \times \frac{7}{5} - \frac{29}{5} \times \frac{12}{5} = \underline{11.08}$$

$$PS = \pi_1 = \frac{98}{25} = \underline{3.92}$$

$$\text{Welfare} = CS + PS = \underline{15}$$

$$\Delta CS = 11.08 - 4 = \underline{7.08}$$

$$\Delta PS = 3.92 - 8 = \underline{-4.08}$$

$$\Delta \text{Welfare} = 15 - 12 = \underline{3}$$

• Problem 3

$$U^j(Q_1^j, Q_2^j) = 12(Q_1^j + Q_2^j) - (Q_1^{j^2} + Q_2^{j^2}) - Q_1^j Q_2^j$$

Demand function of goods produced in i and consumed in j

$$P_i^j = 12 - 2Q_i^j - Q_{-i}^j$$

Input tariff : $t_1 = 1$

Import tariff : $\tau_1 = 4$

Retaliatory tariff : $\tau_2 = 2$

Firm	unit cost sold in Home ($j=1$)	unit cost sold in Foreign ($j=2$)
Home Firm ($i=1$)	$MC_1 + t_1$	$MC_1 + t_1 + \tau_2$
Foreign Firm ($i=2$)	$MC_2 + \tau_2$	MC_2

→ Home firm's ($i=1$) profit :

$$\begin{aligned} \pi_1 &= \pi_1^1 + \pi_1^2 \\ &= \max_{Q_1^1} \left\{ \underbrace{P_1^1 Q_1^1 - (MC_1 + t_1) Q_1^1}_{\text{domestic sales}} \right\} + \max_{Q_1^2} \left\{ \underbrace{P_1^2 Q_1^2 - (MC_1 + t_1 + \tau_2) Q_1^2}_{\text{Export}} \right\} \end{aligned}$$

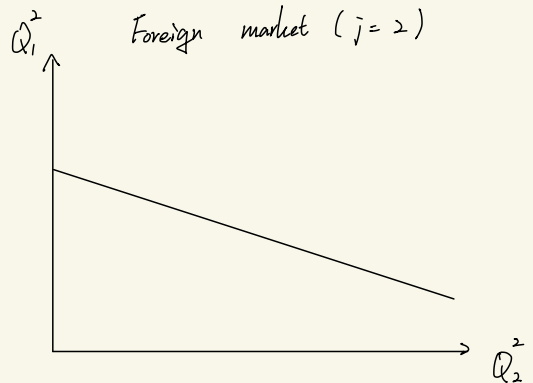
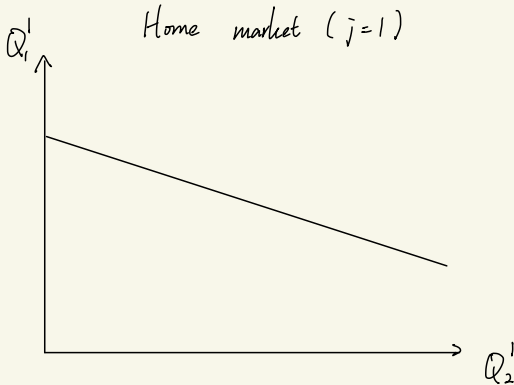
↖ demand function ↖ demand function

$$\text{FOC : } \begin{cases} 12 - 4Q_1^1 - Q_2^1 - (MC_1 + t_1) = 0 \\ 12 - 4Q_1^2 - Q_2^2 - (MC_1 + t_1 + \tau_2) = 0 \end{cases}$$

⇒ Reaction function for Home firm ($i=1$)

$$Q_1^1 = \frac{1}{4} \left[12 - (MC_1 + t_1) - Q_2^1 \right]$$

$$Q_1^2 = \frac{1}{4} \left[12 - (MC_1 + t_1 + \tau_2) - Q_2^2 \right]$$



→ Foreign firm's ($i=2$) profit

$$\pi_2 = \pi_2^1 + \pi_2^2$$

$$= \max_{Q_2^2} \left\{ \underbrace{P_2^2 Q_2^2 - MC_2 Q_2^2}_{\text{domestic sales}} \right\} + \max_{Q_2^1} \left\{ \underbrace{P_2^1 Q_2^1 - (MC_2 + \tau_1) Q_2^1}_{\text{Export}} \right\}$$

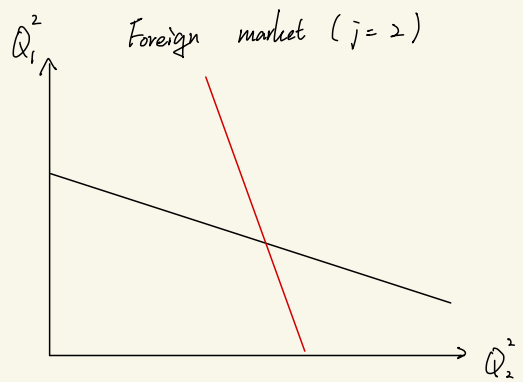
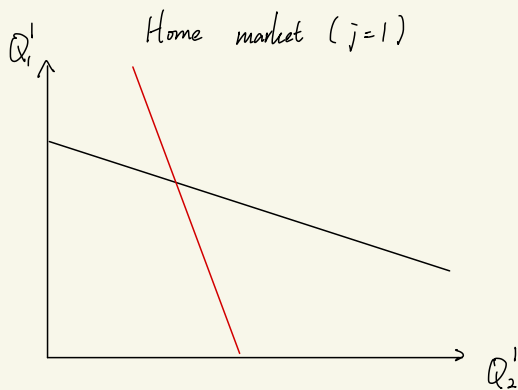
↙ demand function
↙ demand function

$$FOC: \begin{cases} 12 - 4Q_2^2 - Q_1^2 - MC_2 = 0 \\ 12 - 4Q_2^1 - Q_1^1 - (MC_2 + \tau_1) = 0 \end{cases}$$

⇒ Reaction function for Foreign firm ($i=2$)

$$Q_2^2 = \frac{1}{4} \left[12 - MC_2 - Q_1^2 \right]$$

$$Q_2^1 = \frac{1}{4} \left[12 - (MC_2 + \tau_1) - Q_1^1 \right]$$



2. Solve for Q_1^1 , Q_2^1 , Q_2^2 , Q_1^2

$$Q_1^1 = \frac{7}{5} - \frac{4}{15} t_1 + \frac{1}{15} \tau_1 = 1.4$$

$$Q_2^1 = \frac{12}{5} + \frac{1}{15} t_1 - \frac{4}{15} \tau_1 = 1.4$$

↳ free trade
same as Problem 1-2

$$Q_2^2 = \frac{12}{5} + \frac{1}{15} t_1 + \frac{1}{15} \tau_2 = 2.6$$

$$Q_1^2 = \frac{7}{5} - \frac{4}{15} t_1 - \frac{4}{15} \tau_2 = 0.6$$

↳ free trade

4. Domestic Market share & Intra industry trade

$$s_1 = \frac{Q_1^1}{Q_1^1 + Q_2^1}, \quad IIT = 1 - \frac{|X - M|}{X + M} = 1 - \frac{|Q_1^2 - Q_2^1|}{Q_1^2 + Q_2^1} \cdot \frac{1}{1.8}$$

In free trade,

$$s_1^{FT} = \frac{1.4}{1.4 + 2.4} \approx 36.84\%, \quad IIT^{FT} = 1 - \frac{|1.4 - 2.4|}{1.4 + 2.4} \approx 73.68\%$$

With tariffs

$$s_1^{\text{tariff}} = \frac{1.4}{1.4 + 1.4} = 50\%, \quad IIT^{\text{tariff}} = 1 - \frac{|0.6 - 1.4|}{0.6 + 1.4} = 60\%$$